Twitter or WhatsApp? Public and Private Agent Networks Are Different at the Edge

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Abstract

Agent networks are being built to look like Twitter or like WhatsApp, and the two are compared as though "public versus private" were a variable. It is not. An edge is a triple—who may form it, what it carries once formed, what is merely asserted about it—and the two regimes are not two values of one dial but two *bundles* that differ on all three at once: lateral breadth, sandboxed exchange and platform-computed scores against longitudinal depth, openable state and relationship-carried history. Comparing them measures a bundle swap and reports one number that cannot be attributed to any component. We give an instrument that sets the three independently—one model, one runtime, one coordination kernel, conditions expressed as capability grants—and a metric set that prices each axis in accuracy, tokens and disclosure rather than in accuracy alone. Across 582 episodes of hidden-profile coordination scored against forced-value answer keys, the three axes weigh roughly 0.232, 0.080 and 0 on requirement F_1: a length-matched placebo moves the score as much as a trust assertion does, and the formation effect is reach, vanishing when the roster already holds what a task needs. Two results were not designed. A bounded roster is shown a quarter as much misinformation as an open directory and writes more of it into the deliverable, fabricating values 58 times where the open arms fabricated once—filtering out bad claims also removes the second source that catches them. And a single brokered hop recovers $+0.140$ of the gap while every assertion of trust remains flat, which separates an introduction, a formation intervention, from an endorsement, an attribute. The weights are measured on one task family; whether they are constants of the framework or of the task is the experiment the framework now specifies.

1 Introduction

Two pictures of an agent network are in circulation. In the first, agents discover each other in the open: they publish what they can do, broadcast what they need, and are matched by relevance rather than acquaintance. In the second, agents act for people who already work together: membership is bounded, interaction repeats, context accumulates. The first is called public and the second private, and the difference is usually described as one of visibility.

Visibility is the wrong frame, and so is treating the two as endpoints of one dial. Twitter and WhatsApp differ in what it costs to open an edge, in what the edge carries once open, and in what is asserted about the counterpart—three things at once. On the first, an account addresses any other, relevance is a claim in a profile, and a bad actor changes handles. On the second, a conversation exists because a number was already held or someone made an introduction, the tie carries files and history, and misbehaviour costs the tie and embarrasses whoever vouched. Three things differ at once, not one.

Taking the edge as the primitive exposes three things that vary together and are reported as one. Edge formation is which edges a requester may create at all. Edge semantics is what an existing edge carries: a delegated task returning a result, or direct access to state. Edge attributes are the beliefs attached to an edge—a reputation score, a stated shared objective, a note that the parties have worked together before—which change nothing a runtime checks. Public and private are two *bundles* in this space, not two definitions and not two settings of one dial, so a study that compares them swaps all three at once and attributes the difference to whichever component the authors find most interesting. **Public versus private is not the experimental variable**; $do(F)$, $do(S)$ and $do(A)$ are.

This paper unbundles them and prices the components. We place the two regimes as corners of (F, S, A) rather than as definitions (§2), give an instrument that sets the three axes independently and a metric set that charges each one in accuracy, tokens and disclosure (§3), and state the design question the decomposition makes answerable: not which regime wins, but which configuration a setting should hold, and whether the axis weights are constants of the framework or of the task (§4). The coordination kernel is an instrument, not a contribution.

What we have measured so far is one task family, and it is summarised in §5 with the full record—including the nulls and one retraction—in the appendix. Two of its results were not designed and are the reason we think the framework earns its keep. A bounded roster is shown far less misinformation than an open directory and absorbs more of it, fabricating values 58 times against once, because the filter that keeps bad claims out also removes the second source that would catch them. And a single brokered hop—one roster member forwarding a request—recovers a substantial part of the gap, while every assertion *about* trust stays flat. An introduction changes the graph; an endorsement is text on an edge that already exists. The framework predicts that only the first can do work, and that is how it came out.

Contributions. (i) **Public and private recast as two bundles in an edge space** $e = (F, S, A)$ rather than as two definitions, which makes "public versus private" a bundle swap rather than a variable (§2). (ii) **An instrument that sets the three axes independently**—one model, one runtime, one kernel, conditions as capability grants—with a metric set that prices an axis in accuracy, tokens and disclosure rather than accuracy alone (§3). (iii) **A configuration problem in place of a binary comparison**, whose first result is that the space is smaller than it looks and whose sharpest untested prediction is that an edge's position moves (§4). (iv) **A first decomposition on one task family**, with a length-matched placebo on the attribute axis and two undesigned results (§5).

We do not study how a stranger becomes a teammate, and we make no safety claim about either regime. We state in advance that the *existence* of a task-dependent crossing between regimes follows from how we allocate facts and is not a discovery; what is measured is where it falls and what it costs.

2 Public and private are two corners, not two definitions

Take the network as the unit of analysis: agents are nodes, a request from one to another is an edge. An edge is not one thing. It is a triple

$$e = (F,\ S,\ A)$$

— Formation, who may open it; Semantics, what it carries once open; Attributes, what is asserted about it. The three are independently settable, and the first two are checked by a coordination layer on every call while the third only ever reaches a model as text.

> **Public and private are not points on one dial, and they are not definitions we supply. They are two characteristic *bundles* in this space** — two corners that co-occur in deployed systems because each component makes the others affordable. Every real network sits somewhere between them, and most sit at no corner at all.

Twitter and WhatsApp are the two corners. On Twitter an account addresses any other without assent, relevance is a claim in a profile, exchanges reveal nothing but their content, and trust arrives

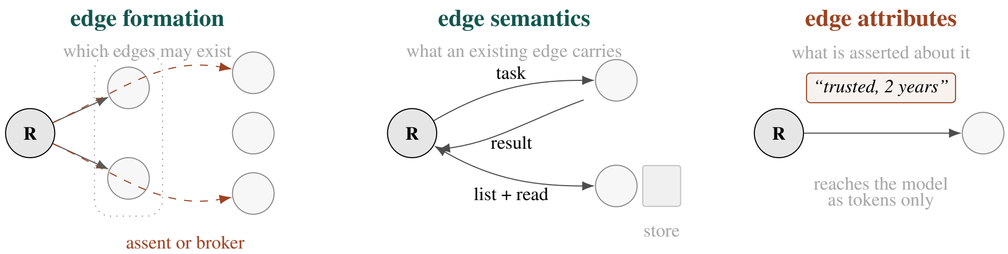

Figure 1: The three axes, drawn on $G = (V, E)$. Formation and semantics are enforced: a coordination layer refuses an edge that may not form and a request an edge does not carry. Attributes are never checked by anything—they arrive in a prompt, which is why asking whether a model acts on them is a real question rather than a tautology. Each regime supplies a characteristic value on all three (Table 1), and a study that compares public with private moves all three at once.

Table 1: The three axes, and what each regime supplies on it. Formation and semantics are enforced by a coordination layer on every call; attributes only ever reach a model as text. A study that compares a public network to a private one moves all three.

	Public	**Private**
Edge formation	Unilateral. Anyone in the directory is addressable; matching is by declared relevance. **Lateral topology**: wide and shallow, many weak ties.	Bilateral or brokered. A counterpart is reachable because of a prior tie or an introduction. **Longitudinal topology**: narrow and deep, few strong ties.
Edge semantics	Sandboxed by default. A counterpart accepts a task and returns a result; its store stays opaque. Interaction is one-shot request/response.	Openable. Shared workspaces, persistent namespaces, a store that may be listed and read—affordable precisely because the tie exists.
Edge attributes	Platform-computed for strangers: reputation scores, ratings, usage counts, verification badges.	Carried by the relationship: tenure, who introduced whom, how the last collaboration went, who is answerable.

as a platform-computed badge; a bad actor changes handles. On WhatsApp a conversation exists because a number was already held or someone made an introduction, the tie carries files and history, what is known about the counterpart is who introduced them and how the last exchange went, and misbehaviour costs the tie and embarrasses whoever vouched.

The consequence is the paper's premise:

> **"Public versus private" is not an experimental variable.** It is a comparison between two bundles that differ on three axes at once. A study that runs it measures $(F_{\text{pub}}, S_{\text{pub}}, A_{\text{pub}}) \to (F_{\text{pri}}, S_{\text{pri}}, A_{\text{pri}})$ and reports one number, which cannot be attributed to any component. The variables are $do(F)$, $do(S)$ and $do(A)$, one at a time.

Table 1 is what each corner supplies on each axis — the bundles unpacked.

Figure 1 draws the three on the graph.The rest of this section takes them one at a time.

Edge formation is which edges a requester may create: the scope of the directory it can enumerate and address. This is the axis the word "public" usually refers to, and the one broadcast networks for agents are built around—an agent publishes what it needs and what it offers, and the network routes to whoever is relevant [EigenFlux]. Which links form, and at what price, is strategic network formation [Jackson and Wolinsky, 1996]; the agent setting differs in that the price is conventionally zero.

The two topologies are the substance of the axis. A public directory buys *breadth*: a rare requirement finds someone, and—less obviously—a claim can be put to a second source. A private roster buys *depth*: fewer counterparts, each already paid for, each with a history. Breadth and depth are not two settings of one dial; they are the returns on two different edge prices.

Edge semantics is what an existing edge carries. A counterpart may accept a task and return a result while revealing nothing else, or expose a store to be listed and read. The distinction is not ours: it separates agent-to-agent task protocols, which deliberately keep a counterpart opaque [Google], from granted resource scopes, in which one principal hands another read access to files. It maps onto the regimes because exposure follows the tie—one does not open a working directory to a stranger—but it is separately settable, and a capability kernel expresses both as grants over different namespaces, so a condition is a set of capabilities rather than a different codebase.

Edge attributes is what is asserted *about* an edge. Both regimes have a characteristic form of it, and both are trust-conveying metadata that no runtime checks: a public network computes a score about a stranger, a private one carries a history about an acquaintance. They rest on the same assumption—that an agent builds a representation of its counterpart from what it is told, and acts differently because of it.

2.1 Why these three and not nine

The literature on ties, teams and open innovation supplies at least nine relational dimensions—membership openness, identity continuity, shared-state depth, repeated interaction, relationship origin, prior trust, shared goals, accountability, exit cost. They are not nine degrees of freedom: in a deployed system an open directory brings weak identity, which brings diffuse accountability, which brings low exit cost. The three-way grouping breaks that coupling the one way a controlled study can. Formation and semantics are properties of the interaction layer—independently settable, checked on every call, not open to argument by a model. Attributes are properties of the people behind the agents; a harness cannot instantiate them, only *assert* them. That asymmetry is the experiment rather than a limitation to apologise for: deployed networks communicate trust, alignment, attribution and tenure as text on cards, so asking whether a model acts on that text asks whether a large part of agent-network design does anything at all.

2.2 Two corrections the framework has to carry

Both fall out of taking the edge, rather than visibility, as the primitive.

A public network's noise is partly a feature. The usual complaint is that an open directory is full of unvetted claims. It is—but a requester who can address many counterparts can put the same question to a second one, and contradiction is *detectable*. A roster that filters misinformation out before it arrives also removes the second source needed to catch what does arrive. Whether the filter or the redundancy dominates is an empirical question, not a definitional one, and the framework must not settle it by calling breadth a cost.

An endorsement is not an introduction. These are on different axes and are routinely conflated. "This counterpart is reliable" is an attribute: text, on an edge that already exists. An introduction that makes a third agent addressable is formation: it changes the graph. A framework that files both under "trust" predicts that a private network's advantage comes from what it knows about its members. The decomposition predicts something narrower—that only the half which changes reachability can do work—and §5 reports which way it went.

3 Making each axis measurable

The framework is only useful if each axis can be set independently and each regime's claimed advantage can be priced. Model, runtime, task, answer key and coordination kernel are held fixed; a condition is a set of capability grants rather than a different codebase, so formation (which cards a requester may enumerate and address), semantics (whether a counterpart returns an answer or exposes a store) and attributes (what the cards say about themselves and each other) move one at a time. Formation and semantics are refused by the kernel when violated; attributes reach the model only as tokens, which is what makes the third axis a test rather than a tautology. Appendix B gives the instrument.

The task class has to force pooling, or reach cannot be worth anything. Each episode is a hidden-profile deliverable: the facts a correct artifact needs are distributed across counterparts so that no single agent, including the requester, holds enough [Stasser and Titus, 1985]. `E`, the share of required facts placed *outside* the roster, is the dose on the formation axis—at $E = 0$ a private network needs nothing it cannot already reach, so the axis must show no effect there. Distractors plant contradicting values on counterparts other than the true holder, so that a second source has something to catch. Every fact is a number, a date, a name or an enum, and the deliverable is scored by exact match after a normalization rule fixed before the keys were written: no judge sits on any primary metric.

An axis is priced, not just scored. Around requirement F_1 sit `useful sources` (counterparts contacted that held a needed fact—the mechanism variable for formation), `pollution absorbed` and `invented` (a distractor's value appearing in the artifact, split by whether the agent was ever shown it, the second being fabrication), and `tokens` and `contacts`—the price of the arm and, since each contact discloses the goal, its exposure. Semantics buys accuracy and is charged in tokens; reporting only the first half would make the private regime look better than the exchange rate supports.

Delivery is reported before any contrast. A trial that produced no artifact is a missing observation, not a score of zero, and averaging it in as zero converts an outage into an effect. We found this the expensive way (§A.6). Every contrast we report is therefore preceded by a per-cell delivery table, is computed both conditional on delivery and with losses scored as zero, and is treated as uninterpretable when delivery differs across compared cells by more than 15 points.

4 From a binary comparison to a configuration problem

Once the bundle is unpacked, the question changes shape. It is no longer which corner wins; it is which point in (F, S, A) a given setting should hold. Three binary axes give eight configurations, and no deployed network is a corner of them: a roster with an introduction protocol is public on formation for whoever a broker can reach and private for everyone else; an enterprise workspace is private on formation and fully open on semantics.

The first result of the decomposition is that the space is smaller than it looks. A configuration search is only a search if more than one dimension moves the outcome. On the task family we ran, the three weights are roughly 0.23, 0.08 and 0 on requirement F_1 (§5), and the third is not small—it is indistinguishable from two neutral lines of text. **An axis with zero weight is not a dimension.** Eight configurations collapse to four, and since semantics costs 11k tokens an episode for a gain whose interval spans zero, four collapse in practice to a choice on formation alone.

That collapse is a finding, not a failure of the framework, and it is the kind of finding only a decomposition can produce: a study that compared the two corners would have reported 0.313 and had no way to learn that one third of the bundle does nothing. We report it rather than restoring the eight-way search, because the honest next question is sharper than an optimisation. **Is the space two-dimensional everywhere, or only here?**

Scenarios are the test of whether the weights are constants. The collapse above was measured on a single task class: facts held by different agents, aggregated into one deliverable. That class is maximally favourable to formation—reach is literally what it asks for—and maximally unfavourable to semantics, since there is no shared artifact to co-edit. If the three weights are properties of the framework, they should hold elsewhere. If they are properties of the task, they will move, and the combination problem becomes real.

The enterprise row is the sharpest test and is deferred: it needs a co-edited task family rather than a reallocation of ours, and we are pursuing it as dedicated work at a venue where the setting can be given the space it needs. The other two rows are stated here as predictions the framework makes. The enterprise row is the sharpest test, because it predicts the reverse of what we measured: when everyone needed is already reachable, formation has no work left to do, and the shared store that cost 11k tokens for a small gain in a pooling task should pay for itself when the deliverable is edited

Table 2: Three settings, and what each predicts for the weight on each axis. These are predictions the framework makes and does not yet have evidence for; the task families are specified in Appendix A.7.

Setting	What a task looks like	formation	semantics	attributes
Enterprise team	Co-edit one artifact; the people are known; the context is the bottleneck	low	**high**	low
Personal errand	Find the one counterpart who can do a specific thing; no relationship needed after	**high**	low	medium
Personal social	Reach a person through someone; the introduction is the product	**high, brokered**	low	**high**

jointly. A framework that only ever produces "formation dominates" is a restatement of the task; one that predicts where formation stops dominating, and is then held to it, is a framework.

4.1 Edges may start public and end private

The framework's sharpest prediction is not about a static configuration at all. If the three axes are separable, an edge's position in (F, S, A) can *move*, and the direction is not arbitrary: each axis unlocks the next.

open discovery $\rightarrow$ sandboxed first exchange $\rightarrow$ repeated success $\rightarrow$ a persistent tie $\rightarrow$ attributes worth carrying $\rightarrow$ shared state

Read against our results this is not speculation about human networks. Public formation is what supplies a second source, and a second source is what our data says catches a wrong claim. Sandboxed semantics is what makes a first exchange with a stranger affordable. Repeated success is the only thing that could make an attribute mean anything—a reputation mark is inert here precisely because nothing generated it. And shared state is priced at 11k tokens, which is affordable only once an edge is expected to be used again. Each stage pays for the next.

This is the framework's most testable prediction and we have not tested it. Every episode we ran is a cold start: 582 episodes at `repeats = 1`. The harness has the knob—a run may repeat a goal family instead of reworking one artifact—and it has never been turned. Two things would falsify the prediction cheaply. If an open requester's contact set does not narrow across repeats, edges do not become private on their own and the transition must be designed rather than emergent. And if the attribute axis stays flat even when the assertions are *earned* rather than stipulated, then $w_A \approx 0$ is a fact about text and not about cold starts.

If instead the contact set does narrow, then "public versus private" is a question about *time* rather than architecture, and a private network is what a public one becomes.

What an optimum means here. An optimum is a configuration, not a size. On formation it is a *funnel depth*: how far past the roster a requester should be allowed to reach—own roster, one brokered hop, two hops, open directory—given that each additional layer widens reach and weakens recourse. The marginal return on a layer is the number of *independent* sources it adds, which saturates; the marginal cost is what a claim from that layer is worth, which decays. The crossing is the depth. Stated this way the optimum is measurable per scenario rather than argued per regime.

5 Evidence so far

We have run the decomposition on one task family: 582 episodes across sixteen sweeps, one model family, one kernel. The full inventory, every result including the nulls and one retraction, is Appendix A. Four findings bear on the framework.

Only formation is an intervention. A naive public-versus-private contrast of 0.313 on requirement F_1 decomposes into 0.232 from edge formation, 0.080 from edge semantics—which spans zero and costs 11k tokens an episode—and nothing distinguishable from zero on attributes. The attribute axis

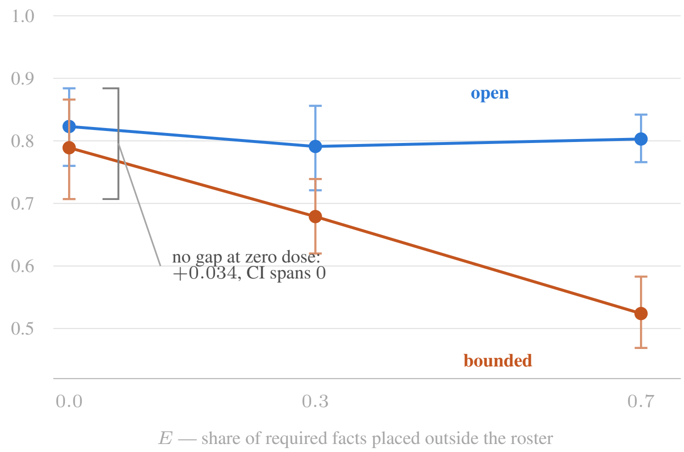

Figure 2: The formation effect is reach, and the dose-response says so. `E` is the share of a task's required facts placed outside the roster; bars are 95% percentile bootstrap intervals over episodes. Four things hold at once, and the axis needs all four: the contrast **vanishes at zero dose**, which is the control it had to pass; the arm that cannot be affected is **flat** across all three doses; the mechanism variable tracks the dose (the bounded arm's useful sources fall $4.36 \to 3.17 \to 1.26$); and fabrication appears only as sources dry up ($0 \to 0.139 \to 0.333$ per task).

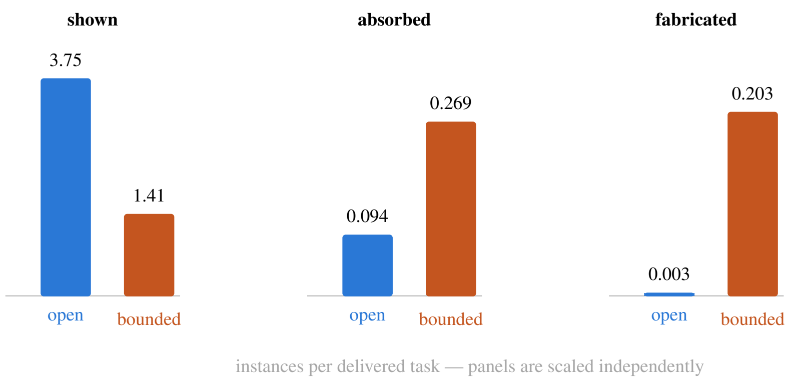

Figure 3: The result nobody designed. A bounded roster is shown roughly a third as much planted misinformation as an open directory—the filter works—and writes almost three times as much of it into the deliverable ($+0.175$, CI $[+0.105, +0.242]$). Worse, when it reaches no one it does not decline to answer: across 287 and 286 delivered tasks the open arms fabricated a value *once* and the bounded arms 58 times. Filtering bad claims out before they arrive removes the second source that would have caught them, so the redundancy argument of §2 shows up here as a failure mode rather than as a benefit.

was tested across 56 contrasts per beat against a chance floor of 2.8; two cleared and reversed sign between beats of the same episodes, and a length-matched placebo of two neutral lines moved the score by $+0.005$. Prior trust and attribution, the two properties deployed reputation systems actually carry, are the flattest measurements in the set.

The formation effect is reach, and the dose-response says so. With `E`—the share of required facts outside the roster—as the dose, the contrast runs $+0.034$ (95% CI $[-0.067, +0.137]$, spans zero) at $E = 0$ (Figure 2), $+0.111$ at $E = 0.3$ and $+0.279$ at $E = 0.7$. The zero-dose cell is the control the axis had to pass and did: a private network that can already reach what a task needs is not worse. The public arm is flat across all three doses ($0.823/0.791/0.803$), so the manipulation is specific to the arm it should affect, and the mechanism variable tracks it—the private arm's useful sources fall $4.36 \to 3.17 \to 1.26$.

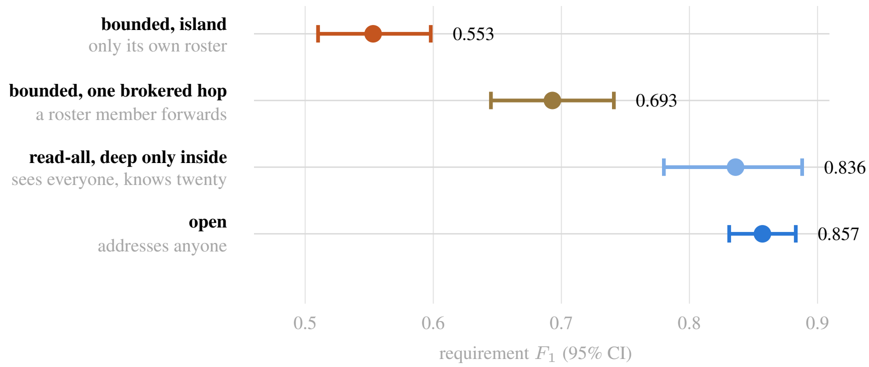

Figure 4: An introduction does work an endorsement does not. All four rows are formation manipulations, and every one of them moves; every attribute manipulation—including the ones asserting exactly the trust a broker would carry—is flat. One brokered hop recovers $+0.140$ (CI $[+0.073, +0.204]$) of the 0.304 gap and cuts fabrication from 0.223 to 0.057 per task; letting a bounded arm read the whole directory while keeping deep relationships inside the roster is indistinguishable from full openness ($+0.019$, spans zero). Depth of relationship is not what is bought here—a second reachable source is.

A roster sees less bad information and believes more of it. This is the one result we did not design (Figure 3). The private arm is shown roughly a quarter as much planted misinformation as the public arm, and writes it into the deliverable *more*—0.269 against 0.094 instances per task (difference $+0.175$, CI $[+0.105, +0.242]$). When it reaches no one at all it does not decline to answer: across matched sets of 287 and 286 delivered tasks, the public arms fabricated a value once and the private arms 58 times. Filtering misinformation out before it arrives also removes the second source that would have caught it, which is the redundancy argument of §2 appearing as a failure mode rather than as a benefit.

That explanation is not the whole mechanism, and we say so because the tidier version is tempting. If missing sources were the only cause, matching on sources obtained should remove the gap; it does not. Among beats that obtained four useful sources, the absorption rate is 2.4% in the open arms and 18.1% in the bounded ones. Starvation amplifies credulity but something else is also operating—plausibly that a counterpart inside a roster is believed *because* it is inside one, which our design cannot separate from the reach effect and which the attribute null makes more interesting rather than less.

An introduction does work an endorsement does not. Giving a bounded roster a single brokered hop—a roster member may forward a request to a counterpart the requester cannot address—recovers $+0.140$ (CI $[+0.073, +0.204]$) of the 0.304 gap to the open arm (Figure 4), replicating independently in three edge-cost strata, and cuts fabrication from 0.223 to 0.057 per task. Letting a bounded arm *read* the whole directory while keeping deep relationships only inside the roster is statistically indistinguishable from full openness ($+0.019$, spans zero). Both are formation manipulations. Every attribute manipulation—including the ones that assert exactly the trust an introduction would carry—is flat. The endorsement/introduction distinction of §2 is the difference between the two halves of that sentence.

What this evidence does not cover. One model family carries every sweep with a live main effect; the replication on a second model was run only at $E = 0$, which is the control cell and is null by construction, so model-independence is not established. No counterpart in these episodes has an incentive to lie: cards may be empty—36 of 100 carry plausible tags and no knowledge—but every agent that is asked answers honestly, so the private regime's characteristic defence has nothing to defend against here. And the scenario predictions of §4 are predictions: we have measured one task family, not three.

6 Further question: the two regimes scale on different bottlenecks

The framework invites a scaling question it does not answer, and unpacking the bundle changes what the question is. The two corners do not grow along the same variable, so they cannot degrade for the same reason.

A public network grows in N, the number of addressable counterparts, and N is not chosen by anyone. Coverage rises with it; so do search cost, verification cost and exposure to counterparts nobody vetted. Its binding constraint is whether ranking can still surface the right counterpart out of N.

A private network cannot grow in N: a roster is bounded by construction. It grows in d, the number of brokered hops a request may travel, because that is the only way trusted coverage increases. Its costs are not retrieval costs—they are the cost of asking a broker to spend its own standing, and the decay of whatever recourse survives a chain of introductions. Its binding constraint is path length and trust transitivity.

> **Public networks scale through ranking; private networks scale through relationships.** The open question the framework poses is not which is better at scale but which degrades faster, under what task distribution, and whether a hybrid inherits the better bottleneck from each.

We can speak to the first of the two and not the second, and only within a range. Growing the directory from 50 to 800 cards—16×, roster held at twenty—leaves the open arm's score flat ($0.800 \to 0.826$, CI spans zero) while search precision falls by roughly half and tokens double: the arm pays for the extra difficulty rather than failing at it. We report this null with a caveat earned the hard way, since our first reading of the same sweep was a scaling result and was an outage (§A.6).

Extrapolating past that range needs a model of how well a directory can be ranked, and the honest version is uncomfortable. A card is ∼50 words of self-description, so what ranking must recover—whether this agent holds *this* value—is a property of private state and is not in the index. Writing ρ for the correlation between rank and actually holding the fact, $\rho = 0$ is not the honest case but the *no-signal* case; web search retrieves from 10^{11} documents, so ranking plainly can be good. The question is how good, and under the model of §A.8 (Figure 6) the answer is that $\rho = 0.9$ has already collapsed by 2×10^5 cards and reaching 10^6 needs $\rho \approx 0.999$. "A better algorithm" understates it: web search reaches its ρ by aggregating outcome feedback rather than by sorting more cleverly, and the equivalent signal in an agent network is a record of who has answered correctly before—which is what a reputation mechanism would have to bootstrap, and our one test of that was null for a reason we can name (§A.5).

The private side is one data point. The brokered hop of §5 is $d = 1$. That single hop recovers $+0.140$ of a 0.304 gap; whether $d = 2$ adds another increment, half of one, or nothing is the shape of the private curve and we have not measured a second point on it. The framework predicts the increments shrink faster than reach grows, because what is spent at each hop—a broker's willingness to attach its own standing to a stranger—is exactly what attenuates with distance while reachable counterparts multiply. If that holds, a private network has an interior optimum in d where a public one has none in N, and the two curves are not two settings of one model.

7 Related work

Agent-to-agent task protocols publish a capability card and accept tasks while deliberately keeping a counterpart opaque [Google]; broadcast networks route published needs among strangers while stating that private information does not enter the network [EigenFlux]; surveys catalogue the wider design space [Agent Protocol Survey, Coral Protocol]. These encode exactly the split we intervene on — whether an edge carries a delegated task or direct access to state — and capability discovery is the first thrust of current agentic-web research, but to our knowledge neither has been evaluated for which side of it changes collective behavior. Which links form, and at what cost, is the subject of strategic network formation [Jackson and Wolinsky, 1996]; our formation axis is that question with the cost set to zero, which is the assumption a follow-up should relax first. The intuition that open networks supply variety and bounded ones supply follow-through descends from weak ties [Granovetter, 1973], exploration and exploitation [March, 1991], transactive memory [Wegner, 1987], and open innovation [Chesbrough, 2003], all studying humans, where the properties in our attribute group are real rather than asserted. Our tasks are the hidden-profile paradigm [Stasser and Titus, 1985], which supplies exact ground truth and makes pooling rather than retrieval the object of measurement; the authority model is object-capability discipline [Miller, 2006], so a condition is a set of grants a kernel checks rather than a convention an agent could ignore. Where an answer key cannot reach, blind pairwise comparison with a cross-family judge is the standard remedy [Zheng et al., 2023], since evaluators favour their own generations [Panickssery et al., 2024]; we preregistered such a protocol but did not run it, and no result here depends on a judge.

8 Conclusion

Public and private agent networks are two bundles of edge properties, not two values of one variable. An edge is a triple—who may form it, what it carries, what is asserted about it—and Twitter and WhatsApp are two corners of that space rather than its two poles. Comparing the corners moves all three axes at once, which is why the comparison has never told anyone which component was doing the work.

Separated on one kernel, one runtime and one model, the three weigh roughly 0.232, 0.080 and 0 on a hidden-profile task family. The formation advantage is reach: it vanishes when the roster already holds what a task needs. The semantics advantage is real but charged at 11k tokens an episode. The attribute axis does not move at all, and a length-matched placebo of two neutral lines is indistinguishable from an assertion of prior trust.

Two results we did not design say more than the ones we did. A bounded roster sees a quarter as much misinformation and writes more of it into its deliverable, fabricating values 58 times where open arms fabricated once: the filter that keeps bad claims out removes the second source that catches them. And one brokered hop recovers a large part of the gap while every assertion of trust stays flat—an introduction changes the graph, an endorsement does not.

The weights were measured on a task family that asks for reach and offers nothing to co-edit. Whether they are constants of the framework or of the task is the next experiment, and it is the one the framework exists to specify: if the enterprise setting does not move the weight on semantics, the combination problem was never a combination problem. For an agentic web the implication is narrow and, we think, useful. Changing what a network permits changes what the collective does. Describing the relationship does not—not at the layer where descriptions currently live.

The decomposition's first result is that the space is smaller than it looks: an axis whose effect is indistinguishable from two neutral lines of text is not a dimension. Whether it is two-dimensional everywhere or only on a task family that asks for reach is the next experiment. So is the framework's sharpest prediction, which we have not tested: that an edge's position is not fixed—that open discovery buys the second source, a sandboxed first exchange makes a stranger affordable, repeated success is the only thing that could make an attribute mean anything, and shared state is worth its price only once an edge expects to be used again. If contact sets narrow across repeats, a private network is what a public one becomes, and the regimes were never alternatives.

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A Everything that was run

Sixteen sweeps, 582 episodes, 2,328 beats, one model family except where noted. This appendix is the complete record: what each sweep varied, what it establishes, and the two that establish nothing. Numbers are recomputed from per-episode summaries over delivered beats, with percentile bootstrap intervals; §3 states why delivery is tabulated before any contrast.

Table 3: Every sweep. "ep" is episodes; each episode is four beats.

Sweep	**Varied**	**ep**	**What it establishes**
`realistic`	$2 \times 2 \times E$	72	Main decomposition: $0.232/0.080/\approx 0$
`axis2x2`	$2 \times 2 \times E$	72	Independent replication of the same
`matched-E0`	three arms, $E = 0$	27	**Zero-dose control: the gap disappears**
`main`	three arms $\times$ E	54	Read-all $\approx$ open; public $-$ private $= +0.240$
`matched-E0-glm`	second model, $E = 0$	18	The control replicates; the main effect is untested
`scaling`	edge cost $\times$ relay depth	112	Brokered hop $+0.140$; per-edge cost is null
`seedaxis`	six attribute profiles	72	Attribute axis flat; placebo $+0.005$
`repscan`	reputation off/real/random	48	Null, but the design is at fault (§A.5)
`capscan`	search cap 5/20/80	24	Reading $16\times$ more results buys nothing
`nscan`	directory 50–800	60	**Retracted** (§A.6)
`k-pilot`	deliverable complexity	8	Pilot only
five smoke sweeps	mechanism checks	15	Confirm each manipulation fires

A.1 Formation: the dose-response and its control

`E` is the share of required facts placed outside the roster.

E	**open**	**bounded**	Δ	**95% CI**	**bounded sources**	**bounded fabrications**
0.0	0.823	0.789	$+0.034$	$[-0.067, +0.137]$ spans 0	4.36	0.000
0.3	0.791	0.679	$+0.111$	$[+0.020, +0.200]$	3.17	0.139
0.7	0.803	0.524	$+0.279$	$[+0.210, +0.345]$	1.26	0.333

Four properties hold together, and the paper's use of this axis rests on all four: the zero-dose contrast spans zero; the arm that should be unaffected is flat ($0.823/0.791/0.803$); the mechanism variable tracks the dose; and the failure mode appears only as sources dry up. The $E = 0$ cells come from the three-arm generation of the harness (`matched-E0`), which is why they are reported alongside rather than inside the 2×2.

A.2 The credulity result

Pooled over `realistic` and `axis2x2`, restricted to delivered beats (287 open, 286 bounded):

	open	**bounded**	**difference**
misinformation shown	$\sim 4\times$ more	baseline	the roster is genuinely cleaner
absorbed into the deliverable	0.094 / task	0.269 / task	$+0.175$ CI $[+0.105, +0.242]$
fabricated outright	1 instance	58 instances	—

`absorbed` requires that the agent was shown the distractor; `fabricated` requires that it was not, so the second is a value invented that happens to coincide with a planted wrong one—a conservative lower bound on fabrication, since only coincidences with planted values are detectable. Stratifying by the number of useful sources obtained does not remove the gap: at four useful sources the absorption rate is 2.4% open against 18.1% bounded, so starvation amplifies the effect but is not the whole of it.

A.3 Introduction against endorsement

From `scaling`, arm C, relay depth 0 against 1, balanced across all three edge-cost levels (28 episodes each):

Architecture	F_1	**useful sources**	**absorption**	**fabrications/task**
bounded, island	0.553	1.50	11.0%	0.223
bounded, one brokered hop	0.693	3.40	6.3%	0.057
read-all, deep only inside	0.836	—	—	0.000
open	0.857	4.63	2.4%	0.000

The hop is worth $+0.140$ (CI $[+0.073, +0.204]$) conditional on delivery and $+0.088$ (CI $[+0.016, +0.158]$) with lost beats scored as zero, and it clears zero independently within each edge-cost stratum ($+0.128$, $+0.138$, $+0.173$). Read-all against open is $+0.019$ and spans zero. This supersedes an earlier estimate of $+0.042$ (CI $[-0.137, +0.208]$) computed from a 16-episode cell; the number in Table 3 is the 28-per-arm one.

A.4 Nulls worth reporting

Per-edge cost does not move the ranking. Charging a handshake turn for each contact outside the roster—194 and 384 handshakes fired at costs of one and two turns—leaves the open arm's advantage at $+0.219$, $+0.263$, $+0.244$ across the three levels while its tokens rise from 53k to 74k. The arm absorbs friction as cost rather than as failure. An earlier draft named this as the variable most likely to reverse the ranking and stated it had not been run; both halves of that sentence are corrected here.

Reading more search results buys nothing. Caps of 5, 20 and 80 give 0.791, 0.755, 0.819, every contrast spanning zero.

Attributes. Six profiles against a length-matched placebo; see §N for the seed texts and §5 for the numbers.

A.5 Reputation: a null caused by the task, not the mechanism

Marks were differential—counterparts that held a planted falsehood were flagged, and a random-mark arm held the wording fixed while removing the information. All three arms are identical: 0.821 off, 0.817 real, 0.833 random, every contrast spanning zero.

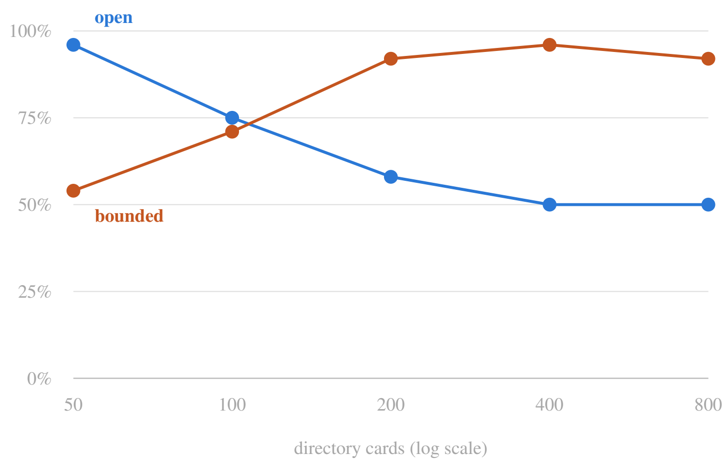

Figure 5: Delivery rate in the retracted sweep: the share of episodes that produced an artifact at all. **The open arm** falls with directory size and **the bounded arm** rises. The bounded arm reads a twenty-card roster at every setting, so directory size is not in its causal path and its delivery rate cannot be responding to it. Both curves are the same transport outage, sampled by two arms that happened to walk their size settings in opposite orders. Scoring the missing artifacts as $F_1 = 0$ turned this figure into a scaling result.

The design is at fault. Each fact has exactly one holder, so a flagged counterpart is also the only route to the true values it holds; a requester that avoided flagged cards would lose more than it gained, and routing measurement confirms it moved *toward* them. Making the mark actionable requires two holders per fact, one reliable and one not, which is a change to fact allocation rather than to the reputation mechanism. We report this as a failed manipulation rather than as evidence about reputation.

A.6 The retracted directory-size sweep

`nscan` varied the directory from 50 to 800 cards and appeared to show the open arm's advantage decaying to zero. It does not. 27% of its beats produced no artifact, every one of them ending in a transport failure after four retries, and the scorer's $F_1 = 0$ for a beat with no artifact was averaged in as a score. Other sweeps lost 2–3%.

Three checks establish that the losses are not caused by directory size. The bounded arm reads a twenty-card roster at every setting, so size is not in its causal path, yet its delivery rate rose from 54% to 96% across the sweep (Figure 5). Scenario was collinear with wall clock—one scenario ran entirely inside the failure window—and the two arms walked their size settings in opposite orders inside it, which is what drew one line down and the other up. And lost beats used 4.4 of 22+ available turns with zero build calls while taking *longer* in wall clock, which is a stalled connection rather than an exhausted budget.

Conditional on delivery both arms are flat: the open arm runs $0.800, 0.777, 0.758, 0.779, 0.826$ across a $16\times$ range ($\Delta = -0.026$, CI $[-0.123, +0.065]$). In the one cell with full delivery at every size, search precision falls from 0.528 to 0.386 while the score holds and tokens double—the finding reported in §6, at $n = 3$ episodes per point and therefore as a hypothesis for a rerun.

The analysis code now computes per-cell delivery before any contrast and refuses to interpret one when delivery differs across compared cells by more than 15 points.

A.7 Task families for the scenario test

The scenario predictions of §4 require three task families rather than one. Each keeps forced-value scoring and hidden-profile allocation; what changes is where the difficulty sits.

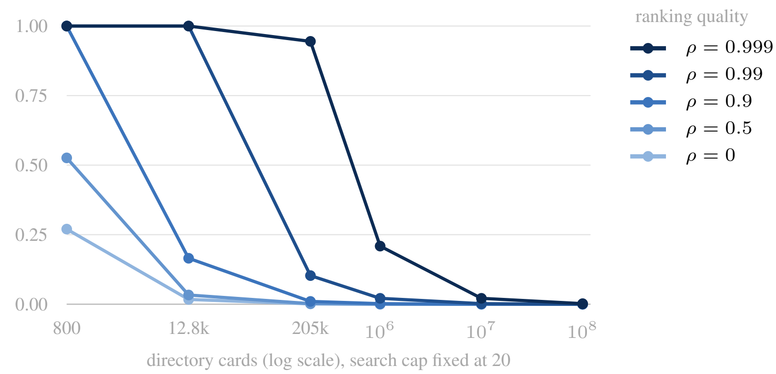

Figure 6: How good must ranking be for a public directory to keep working. ρ is the correlation between a card's rank and whether its owner actually holds the needed fact; the vertical axis is the probability that the holder survives truncation at 20 results. $\rho = 0$ is not the honest case but the *no-signal* case—web search retrieves from 10^{11} documents, so ranking plainly can be good. The question is how good, and the answer is uncomfortable: $\rho = 0.9$ has collapsed by 2×10^5 cards, and reaching 10^6 needs $\rho \approx 0.999$. "A better algorithm" understates it; web search reaches its ρ by aggregating outcome feedback, which is what a reputation mechanism would have to bootstrap (§A.5).

- **Enterprise team.** One artifact revised across four beats by counterparts who all hold part of the context and all remain reachable ($E = 0$). Formation is starved by construction; the shared store is the only route to the accumulated state. This is the row that predicts the reverse of what we measured.
- **Personal errand.** A single specific capability is needed once, held by one counterpart among many that claim adjacency, with no follow-up beat. Formation should dominate more sharply than in the pooling family, and semantics should be worthless.
- **Personal social.** The deliverable requires reaching a counterpart who is addressable only through a chain of brokers, with each hop's willingness a function of what the previous hop asserts. This is the only family in which an attribute is load-bearing, because it gates whether the next edge forms at all—which, if it holds, would locate the attribute axis's one real use.

A.8 The retrieval model behind the scaling question

Let k be the number of search results a requester acts on, m the number of cards matching a query, h the number of those that actually hold the needed fact, and ρ the correlation between a card's rank and holding it. With no ranking signal the needed holder's expected rank is $(m+1)/2$; with perfect ranking it is $(h+1)/2$. Interpolating between the two by ρ and truncating at k gives the survival probability plotted in Figure 6. We take $m(N) \approx 0.094N$ from the directory generator, which draws 35% of filler tags from the scenario vocabulary, and hold $k = 20$.

The model is deliberately crude—it assumes a single query and ignores reformulation—and its purpose is not to predict a number but to establish an order of magnitude for the requirement on ρ. That requirement is the load-bearing claim: at 10^6 cards the ranking has to be right to within one part in a thousand, which is not a threshold a better similarity function reaches.

B Interventions on the graph

Formation and semantics are interventions on different parts of $G = (V, E)$, and Figure 7 exists to keep them apart. Formation changes which edges a requester may create: open discovery lets it address any of 100 holders, bounded discovery restricts it to a roster of 20. Semantics changes what an existing edge carries: e_{ij} either delegates a task and returns a result, or grants `list` and `read` over a store.

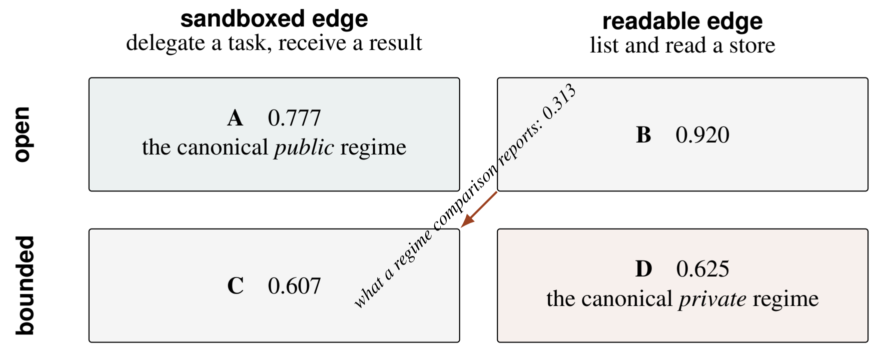

Figure 7: Requirement F_1 at the cold beat. Comparing only the diagonal confounds formation with semantics. B and C are the controls that recover both main effects: $\Delta_{\text{formation}} = \frac{1}{2}[(A - C) + (B - D)] = 0.232$ and $\Delta_{\text{semantics}} = \frac{1}{2}[(B - A) + (D - C)] = 0.080$, summing to the 0.313 a regime comparison would report as one number. C is not a contrivance: a trusted counterpart that answers rather than exposes is the common case inside organizations.

Two further dimensions ride on the same mechanism and are set with the cell rather than varied independently: identity continuity is whether a counterpart's namespace survives the encounter, and repeated interaction is whether later beats reuse it. Delegation depth is bounded per cell and refusals are logged, so a counterpart that tries to sub-delegate past its ceiling produces an event rather than a silent success. The kernel is an instrument: it lets us set properties deployed systems couple together, and it refuses an agent that tries to exceed them.

C Seeding the textual dimensions

Appendix N gives the seed text and the behavioural readout for each profile. The five attribute dimensions are properties of the people behind the agents, and a harness cannot instantiate them. It can assert them—which is exactly what a deployed agent card, reputation field, or relationship summary does. We therefore seed each as context and measure whether behavior changes.

The origin condition is the cleanest instrument. Four seeds change what is stated about incentives or consequences. RELATIONSHIP ORIGIN changes only how the encounter is narrated: same directory, same grants, same knowledge, same store. If the primary metrics move under that manipulation alone, relational framing does real work; if they do not, richer metadata on a card is unlikely to do better, because it is the same channel carrying more words.

Validity boundary. *Seeded trust is not trust.* We measure whether an agent acts differently when told it is in a trusting, aligned, accountable or long-standing relationship—the sensitivity of a runtime to asserted relational context, which is what an agent network actually delivers to a model. We do not measure what trust does in a human network, and no claim here should be read that way. If seeds move behavior the paper reports elasticities usable in card and protocol design; if not, it reports that relational metadata is inert without architectural backing.

D Tasks and measurement

D.1 Deliverables with forced values

Every task is a hidden-profile artifact. The facts a deliverable requires are split across specialist counterparts, and plausible-but-wrong *distractors* are planted alongside them—each on a counterpart other than the holder of the value it contradicts, and at least one inside the bounded roster, so no configuration is immune by construction. Ground truth is designed rather than inferred, so scoring is set arithmetic.

Two schema properties do the work. Every required field is *forced*: its value is determined by the facts, so no field can be a matter of taste. And one fact set has two projections—a flat *report* whose fields are values counterparts hold, and a *chained decision* whose fields depend on one another—which makes coordination depth a real dimension rather than a chain welded onto a flat task. Elimination forces pooling: one constraint rules out one option, a second another, a third a third, each on a different counterpart, so the surviving choice is unreachable from any one of them. Tasks also carry a *craft* component, a rendered artifact with structural requirements, because a value can be retrieved in one message while a working artifact cannot—which is where iteration and delegation depth begin to matter. Three scoring rules are fixed before any key is written (Appendix I): the schema and its enumeration vocabulary are given to the agent so only values are unknown, normalization is specified in advance, and evidence is matched on values rather than internal identifiers.

D.2 Metrics

Five metrics are exact-match or counters and carry the paper's claims: `requirement_f1`, the F_1 over required values against distractors asserted; `completion`, schema conformance used as a gate; `realization`, achieved recall divided by the recall attainable given the configuration's reach; `boundary_cost`, the sanitize, filter, verify and distil actions with their tokens; and `crossing_count`, how many times a hybrid episode changed regime.

`realization` is what keeps the study honest. The allocation fixes how much any configuration *could* obtain; dividing it out leaves what the configuration converted—whether it found the right counterparts, asked enough of them, and resolved the conflicts. The ceiling is designed; the realization is not. Two instrumentation rules follow from a defect we want to make impossible: completion is decided by the harness's deterministic checker and never read from a condition-specific field—such a field exists on the private path of the deployed system we draw from and has no public counterpart, so using it would let "private completes more" prove itself—and token accounting is counted in the harness for the same reason.

Three quantities an answer key cannot reach—overall utility, novelty, and diversity of approach—were preregistered for blind pairwise scoring by a cross-family judge, with agreement reported. **They were not run for this submission.** No result in this paper depends on a judge, and we list them as planned work rather than describe a protocol we did not execute.

Honesty statement. The allocation fixes how much each configuration can reach, so the *existence* of a task-dependent crossing is a property of the design and not a discovery. What is measured is where it falls, what it costs, and how much of the attainable each configuration realizes. The seeded conditions and the access axis carry no such guarantee, and they are the study's primary empirical content.

E Experimental design

E.1 Predictions

Each prediction was chosen against one filter: *would a careful reader bet on it in advance?* Claims that pass trivially—an open network reaches more knowledge, a persistent team completes more—are excluded, because confirming them consumes a study and returns nothing. Every prediction below was fixed before the corresponding sweep ran, together with the condition that would falsify it.

H3 is worth naming as a failure. We predicted that a counterpart answering from its own full context would beat a requester re-joining fragments it had read—that extraction is lossy in a way delegation is not. On accuracy it is not: readable edges are ahead in both arms. The prediction survives only once cost is counted, which is a different and weaker claim, and we report it as H3$'$ rather than presenting the original as confirmed.

E.2 Conditions

Four cells of the 2×2, each a configuration of the same kernel: open or bounded edge formation crossed with sandboxed or readable edge semantics. Cells B and C are what allow the two main effects to be identified rather than summed; without them, an open-versus-bounded contrast reports $\Delta_{\text{formation}} + \Delta_{\text{semantics}}$ and cannot say which term carried it.

Table 4: Predictions, fixed before the runs, with the outcome. Two survived, one was falsified in a way that changed the paper's second finding, and one was withdrawn.

	Prediction	Falsified if
H1	Edge formation dominates edge attributes	attribute contrasts move outcomes comp rably to formation
H2	Attributes are inert once text length is controlled	a relational seed beats a length-match placebo consistently
H3	Sandboxed edges beat readable ones on accuracy	readable edges are at least as accurate
H3′	Sandboxed edges win *per token*	readable edges are more efficient per tok in either arm
H4	The formation advantage is a reach advantage, and shrinks when reach stops being scarce	the gap is flat in E and across beats
H5	Readable edges absorb more misinformation	absorption tracks exposure alone

Table 5: Requirement F_1 at the cold beat, by cell, and the two main effects. The naive regime contrast is 0.313; three quarters of it is who can be reached and one quarter is what crosses the edge.

	sandboxed	readable
open	0.777	0.920
bounded	0.607	0.625

$\Delta_{\text{formation}} = 0.232$ 95% CI $[0.127, 0.316]$ excludes 0
$\Delta_{\text{semantics}} = 0.080$ 95% CI $[-0.011, 0.175]$ spans 0

The attribute axis holds the configuration fixed and varies only what is asserted in context, across five relational profiles and a length-matched placebo.

E.3 Controls

One model across all agents and all cells, one runtime, one kernel; identical task prompts, deliverable schemas, scoring functions, token budgets and seed distributions. Rate limits and loop bounds are logged per episode rather than absorbed into results. Rubrics, schemas, normalization rules and the scorer signature were frozen before any run (Appendix J).

Two controls deserve naming because each removed an explanation that would otherwise have survived into the paper. The **requester keeps its own notes in every cell**: withholding them from the open configurations would have manufactured the warm-start result, so what differs is not memory but whether memory is bilateral. And the **length-matched placebo** carries the same line count, position and register as the relational seeds with no relational content, which is what separates an effect of the relationship from an effect of the tokens describing it.

F Results

All numbers below come from 144 episodes on one kernel, one runtime, and one model: 72 for the 2×2 over edge formation and edge semantics, 72 for the attribute axis. Bootstrap intervals are percentile over episodes, resampled within cell.

F.1 A naive public-versus-private comparison, decomposed

The contrast a study of "public against private" would report is open+store against bounded+sandbox: 0.313 on requirement F_1 at the cold beat. It decomposes almost exactly additively into the two interventions.

F.2 Edge formation dominates (H1)

Across seven tests—four accuracy metrics and three cost metrics, over the cold, rework and warm beats—$\Delta_{\text{formation}}$ clears its interval in six. Chance at $\alpha = 0.05$ would produce 0.4. The sign is stable and the magnitude is largest where the task most needs outside knowledge: at a high external fraction it reaches 0.352, against 0.112 when most holders sit inside the roster.

Table 6: Attribute contrasts on requirement F_1, stratified by (arm, scenario) over four strata, $n = 12$ per profile. The placebo matches the relational seeds in line count, position and register while carrying no relational content.

contrast	**cold**	**warm**	
placebo − control *added text alone*	$+0.005$	-0.014	spans 0
trust − control	-0.010	$+0.017$	spans 0
attribution − control	-0.046	$+0.015$	spans 0
colleague − placebo	-0.088	$\mathbf{+0.074}$	sign flips
stranger − placebo	$\mathbf{-0.116}$	$+0.042$	sign flips
colleague − stranger *narration only*	$+0.029$	$+0.032$	spans 0

It is also cheaper. Opening the directory saves 17.1k tokens per episode (CI excludes 0) while raising accuracy, because a requester that can address the right holder stops paying for the search around it.

F.3 Edge semantics: a small gain at a real price (H3)

$\Delta_{\text{semantics}}$ clears in two of the same seven tests, against a chance floor of 0.4. The sign is positive in all three beats (0.080, 0.028, 0.098) but only the warm beat's interval excludes zero, so we report it as small and underpowered rather than absent.

The cost side is unambiguous, and it inverts the ranking. Reading a counterpart's store costs 11.0k tokens per episode (CI excludes 0). Per token spent, agent mediation is ahead in both arms:

	sandboxed	readable	ratio
open	0.177	0.157	1.13×
bounded	0.094	0.087	1.08×

F_1 per 10k tokens. Exposing state buys a little accuracy; delegating to an agent that already holds the context buys more of it per unit of context spent. This is the tradeoff between moving data to the computation and moving the computation to the data, measured on an agent network.

F.4 Edge attributes are inert (H2)

The attribute axis asserts five relational properties in context—prior trust, attribution, a two-year colleague framing, a stranger framing, and a length-matched placebo—with the network configuration held fixed.

Two results matter here. First, the placebo settles the alternative explanation: two lines of neutral text, matched for length and position, cost $+0.005$ with an interval spanning zero. Whatever the relational seeds do, they do not do it by occupying tokens.

Second, nothing survives. Eight contrasts across seven metrics is 56 tests per beat, so chance alone yields about 2.8; two cleared, and they cleared *in opposite directions in different beats of the same episodes*. The narration instrument—the cleanest of the set, differing in one sentence and nothing else—is flat at 0.029. The two assertions that deployed reputation systems actually implement, prior trust and attribution, are the flattest of all.

F.5 Summary

intervention	**effect**	**tests cleared**	**cost**
edge formation	0.232	6/7 (floor 0.4)	-17.1k tokens
edge semantics	0.080	2/7 (floor 0.4)	$+11.0$k tokens
edge attributes	≈ 0	2/56 (floor 2.8), signs flip	$+0.005$

G Discussion and limitations

G.1 The formation advantage is a reach advantage

$\Delta_{\text{formation}}$ is not a constant. It tracks how much of what a task needs sits outside the roster, and it collapses when reach stops being scarce.

At a low external fraction the gap is a third of its high-E value ($+0.112$ against $+0.352$ cold), and on the rework beat—where the task must return to a counterpart it has already used—the point estimate reverses to -0.037: the bounded network is ahead. Averaged over beats, the formation effect halves on rework ($0.232 \rightarrow 0.114$).

We report the reversal as suggestive rather than established: it is a subcell point estimate and we do not attach an interval to it. But the pattern across the table is consistent and it bounds the headline. The open network wins because it can address holders the bounded one cannot; where the bounded roster already contains the holders, and where the work requires going back to them, that advantage is largely spent.

G.2 A bounded roster as a wall, and what happens when it is not

A bounded network here addresses twenty holders and no others, which is a truncated directory rather than a private network: information reaches a closed group through the people in it, one hop further out. We let a roster member forward one question to a contact of its own, minted under its own authority, and re-ran. **The gap did not closes about half**: the hop is worth $+0.140$ (CI $[+0.073, +0.204]$) over 28 episodes per arm, replicating independently within each of the three edge-cost strata, and it cuts fabrication from 0.223 to 0.057 per task. A first estimate from a 16-episode cell put it at $+0.042$ with an interval spanning zero; §A.3 gives the current numbers and §P the design and the harness failure it exposed.

G.3 Forming an edge is free here, and that is the main limitation

Whether a link is worth its cost is the central question of strategic network formation [Jackson and Wolinsky, 1996], where agents weigh the benefit of a connection against the price of maintaining it. **We set that price to zero.** In this design, addressing a stranger costs exactly what addressing a teammate costs. The requester searches a directory and sends a message; there is no negotiation, no context transfer, no format alignment, no establishing what a counterpart can actually do.

This removes the economic reason relationships exist. An open arm with a hundred addressable holders at zero acquisition cost is closer to an oracle than to a network, and the formation effect we measure should be read as an upper bound: it is what open discovery is worth *when discovery is free*. A per-new-edge cost—an onboarding exchange, a latency penalty, a token charge on first contact—is the single variable most likely to move the ranking. We have since run it, and it does not: charging one and then two handshake turns per contact outside the roster—578 handshakes fired—leaves the open arm ahead by $+0.219$, $+0.263$ and $+0.244$ across the three levels while its tokens rise from 53k to 74k. The arm converts friction into cost rather than into failure, which is the same shape as the directory-size null (§A.6). A cap on *total* contacts, which cannot be paid off by spending more, is the manipulation that remains untried.

G.4 What the attribute null does and does not bound

The null covers assertions in a prompt, the weakest form of relational encoding. It says nothing about trust wired into routing, memory, retrieval or permission policy, where reputation lives in deployed systems and where it changes what an agent is able to do rather than what it is told.

It is narrower still, in a way we can name: each seeded property had *no available action* behind it. Trust and origin were asserted uniformly over every card, ranking no counterpart above another and so unable to change who gets asked first — deployed reputation is differential, and that is the whole of its value. Accountability went to responders whose knowledge is a fixed list, leaving them nothing to do differently.

A fourth manipulation removes the first defect: a per-card line reporting how often that card's past answers held up, accurate by construction, against a placebo assigning the same marks at random.

The marks are perfect — 100% accurate against 9% — and F_1 still does not move (-0.017, CI $[-0.174, 0.143]$). The routing measurement says why: under the accurate marks the requester contacts flagged cards at *six times* the base rate rather than avoiding them, because the flagged cards hold the planted falsehoods and in this task those are the same cards holding the true facts on their subject. Acting on the flag meant not asking the only source.

So the claim is not that relational metadata does not work, but that it moves behavior only where it discriminates between options an agent can choose between — and three signals that did not, one uniform, one aimed at an agent without discretion, one flagging the sole source, moved nothing. Making it actionable needs two holders per fact, one reliable and one not: a change to the task, not to a configuration.

Four further limits, stated plainly: the cold-phase crossing follows from the fact allocation and is not a discovery, so only its location and cost are measured; forced-value deliverables cannot capture whether an artifact is well argued, which is the price of removing the judge; the kernel is a reference implementation rather than a deployment; and one model family carries every sweep.

H Hypothesis-to-figure map

[TODO: Frozen before runs; reproduced here so the analysis plan is auditable.]

I Scoring rules fixed before key authoring

Three rules are fixed before any key is written. The schema—field names, types, and the full enumeration vocabulary—is given to the agent, so that only values are unknown and we are not measuring whether a model guesses our vocabulary. Normalization is specified in advance: numbers parsed and compared exactly, dates ISO-8601, strings casefolded and trimmed, enumerations restricted to the supplied list, lists compared as sets. Evidence is matched on values rather than on internal identifiers, which agents never see.

J Controls and preregistration

One model across all agents and all cells; one runtime; one kernel. Identical task prompts, deliverable schemas, scoring functions, token budgets, and seed distributions. Rate limits and loop bounds are logged per episode rather than absorbed into results. Rubrics, schemas, normalization rules and the scorer signature are frozen before any run; the mapping from each hypothesis to the figure that tests it is fixed in advance and reproduced in Appendix H.

K Threats to validity

A null on the textual group could be a prompt-engineering artifact rather than a property of the runtime: a stronger seed, placed differently in context, might move behavior. We mitigate by seeding at three intensities and reporting the dose-response, and by including the RELATIONSHIP ORIGIN condition, which is the weakest possible seed and therefore the cleanest test of whether framing alone does anything. A null could also be a ceiling effect if the tasks are too easy to differentiate configurations; `realization` is reported partly to detect this.

L Deliverable schemas and normalization rules

[TODO: Both projections of each scenario; the full enumeration vocabularies handed to agents.]

M Directory construction

[TODO: Card counts by group (fact-bearing, near-miss, noise), the shared tag vocabulary, and the roster compositions realizing each level of external fraction.]

N Seeding protocol

Table 7: Seeding protocol. Each dimension is asserted as context in the seeded condition and omitted in the control; everything else, including the architectural configuration, is held fixed. The readout column names the behavioral quantity that should move if the seeded property is doing any work.

Dimension	Seed	Readout
PRIOR TRUST	Requester memory is pre-populated with an interaction record: three prior exchanges with a named counterpart, all accurate.	Verification actions per obtained value; cross-checks against a second holder.
SHARED GOALS	The counterpart's context states the requester's objective as its own, versus a divergent one (bills per answer; minimizes its own effort).	Answer completeness; unsolicited volunteering; refusal rate.
ACCOUNTABILITY	The counterpart is told its answers are logged, attributed, and that accuracy is written back to a record others read, versus anonymous and unlogged.	Error rate; rate of explicit "I do not hold that".
EXIT COST	The counterpart may decline or fall silent without consequence, versus being obligated to respond.	Decline rate; retries the requester must spend.
RELATIONSHIP ORIGIN	Narrative framing only: "you found this agent on an open board" versus "this is a teammate of two years". *Nothing else differs.*	Any movement at all in the primary metrics.

[TODO: Verbatim seed text for each profile.]

O Reproducibility

[TODO: Kernel version, runtime version, model identifiers, seeds, and the command that regenerates every table and figure from the run logs.]

P Relay: a bounded roster given a second hop

A bounded network here addresses twenty holders and no others, which is a truncated directory rather than a private network: information reaches a closed group through the people in it, one hop further out. We therefore let a roster member forward one question to a contact of its own, minted under its own authority rather than derived from the requester's, and re-ran. **The gap did not close.** Across 64 episodes at sixteen per cell relay moves the bounded arm by $+0.042$ (CI $[-0.137, 0.208]$), and the difference-in-differences against the open arm — whose configuration is identical under both settings and which serves as the negative control, moving -0.063 — is $+0.105$ (CI $[-0.105, 0.322]$).

Two details survive it. The same contrast read $+0.255$ at six episodes per cell with an interval excluding zero and decayed monotonically with n. And excluding episodes that produced no artifact, the bounded arm scores 0.734 with relay against 0.554 without: relay helps substantially *when it completes* and costs enough exploration budget that seven episodes never reached a build — five under relay and two under a high first-contact cost without it, so the failure belongs to any mechanism that consumes extra turns, and exposes a harness that answers budget exhaustion by refusing tools rather than by forcing a build.